Math Cat

- The cat asks you a **sum**.
- Type the answer and press **Enter**.
- Every right answer, the cat **walks** across.
- Get **10 right** to win.

> Build it yourself in one hour. Every mission needs a green **Operators** block.



Set up

1. Go to **scratch.mit.edu** and click **Create**.
2. **Keep the cat.** Everything today goes on the cat (Sprite1).
3. Click **Variables**, then **Make a Variable**. Name it **Score** and click OK.
4. Make two more the same way: **A** and **B**.
5. Click the green **Operators** section and look at the blocks. Every mission uses one.

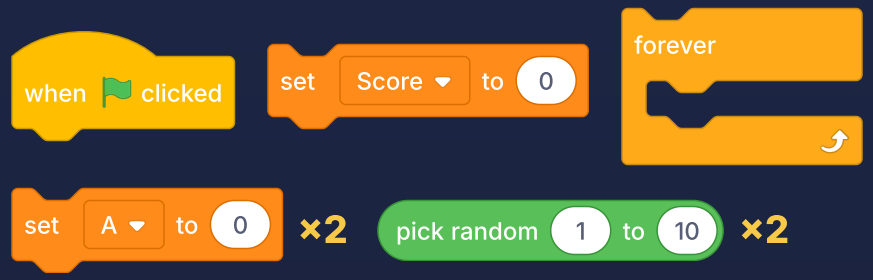
> Capital letters: **Score**, **A** and **B**. In each orange block, click the little arrow to pick the right variable.

Cat

1 · Pick

When the green flag is clicked: set Score to 0. Then, forever: set **A** to **pick random 1 to 10**, and set **B** to pick random 1 to 10.

BLOCKS YOU NEED



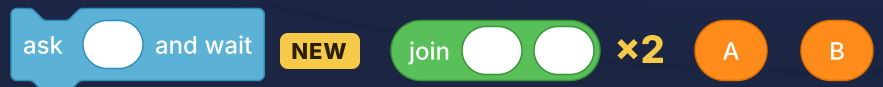
Test it: click the green flag. A and B change super fast on the stage.

Cat

2 · Ask

Inside the forever, under the two sets: **ask** the sum and wait. Put a **join inside a join**: A, then **“ + ”** (a space each side), then B.

BLOCKS YOU NEED



Test it: the cat asks a sum, like 7 + 5. Type anything and press Enter.

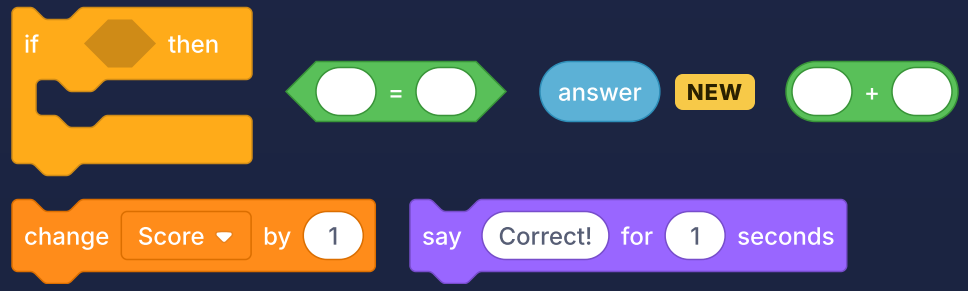
Click the green flag after every mission. Stuck? Try for **3 minutes**, then raise your hand.

Cat

3 · Right!

Under the ask: if the **answer = A + B**, change Score by 1 and say Correct! for 1 second.

BLOCKS YOU NEED



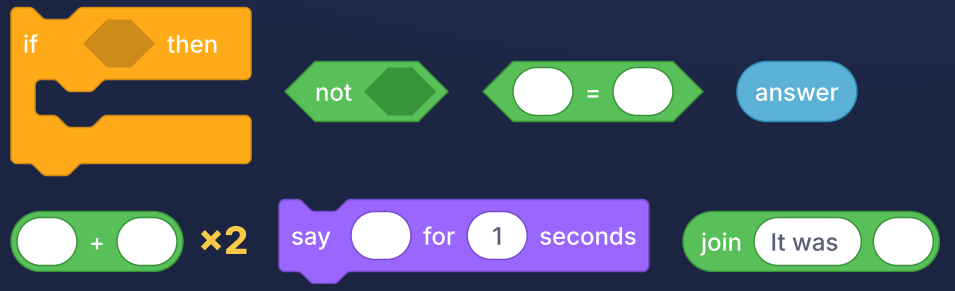
Test it: answer right. Score goes up by 1.

Cat

4 · Wrong

Under that if, a second if: if **not** (answer = A + B), say **join** “It was ” and A + B, for 1 second.

BLOCKS YOU NEED



Test it: answer wrong. The cat tells you the right answer.

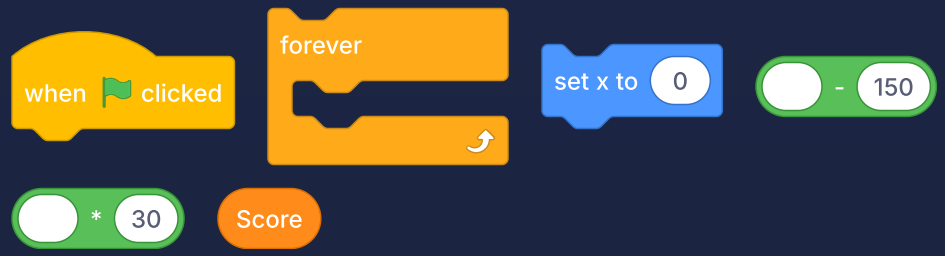
Click the green flag after every mission. Stuck? Try for **3 minutes**, then raise your hand.

Cat · a new script

5 · Walk

A second script on the cat. When the green flag is clicked, forever: set x to **(Score * 30) - 150**. Put the ***** block inside the **-** block.

BLOCKS YOU NEED



Test it: every right answer, the cat walks 30 steps right.

Cat · the first script

6 · Win

At the bottom of the first script's forever: if **Score > 9**, say You win! for 2 seconds, then stop all.

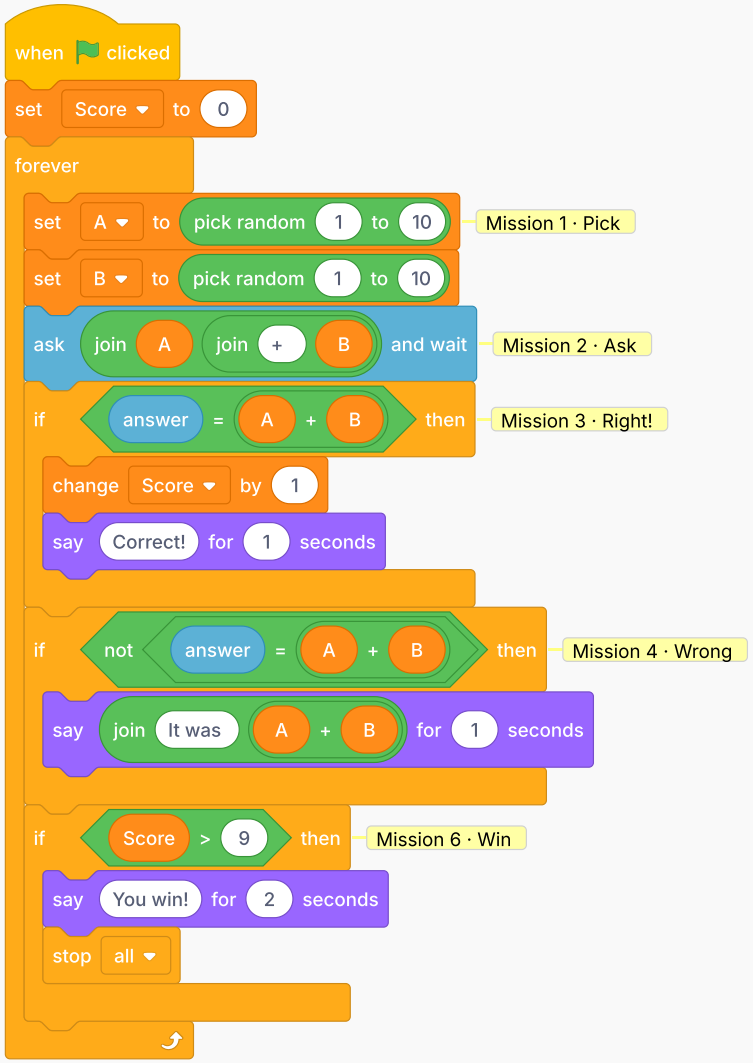
BLOCKS YOU NEED

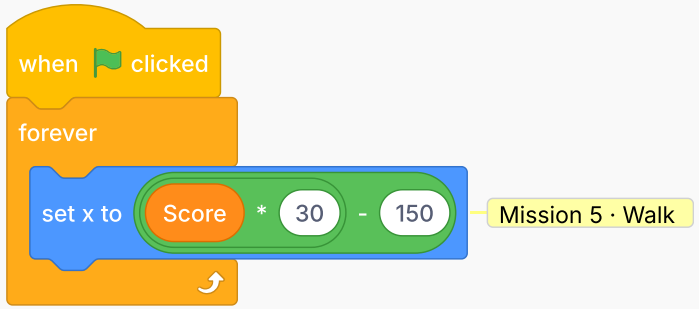


Test it: get 10 right. The cat wins and everything stops.

All six work? Try these: pick random 1 to 20 · make it times: * instead of + in all three places, and " x " in the join · start sound Meow when you're right · a 60-second countdown on the Stage, like Shark Chase

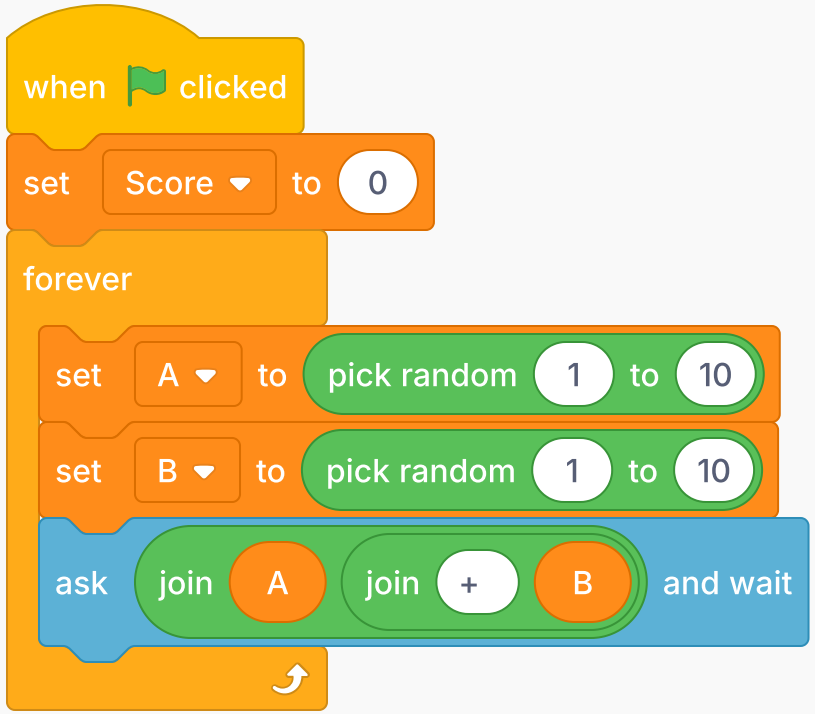
The finished code





For the teacher: this is the whole game. Both scripts go on the cat (Sprite1), and each starts with its own green flag. The next three slides show each part up close: go through them one at a time, and ask the class what each green block does. Then everyone fixes their own game.

Missions 1 and 2: pick and ask



pick random a new number from 1 to 10, every time round the loop. A and B are different every question.

join glues two things into one. The inside join makes **" + 5"**. The outside one puts A in front: **"7 + 5"**.

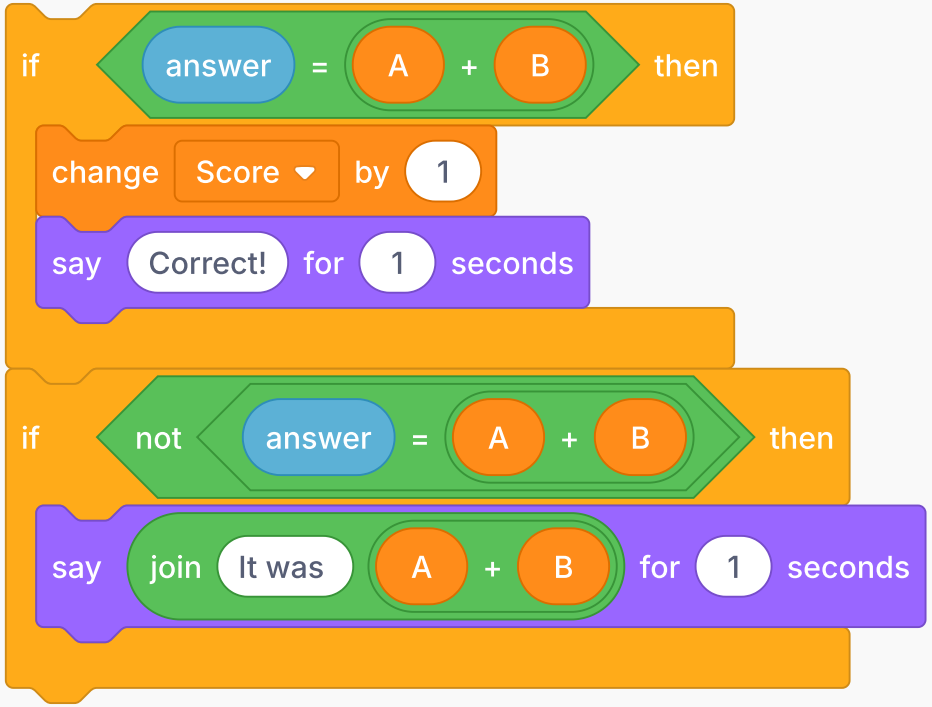
If it doesn't work

The question reads 75: type **" + "** into the inside join, with a space each side.

A and B never change: the two sets go **inside** the forever.

Missions 3 and 4: right or wrong

These two ifs go inside the forever, right under the ask.



+	adds A and B. That's the right answer.
=	asks: is what you typed the same as A + B?
not	flips it: **not** the same means the answer was wrong.
join	again: “It was ” and the right answer.

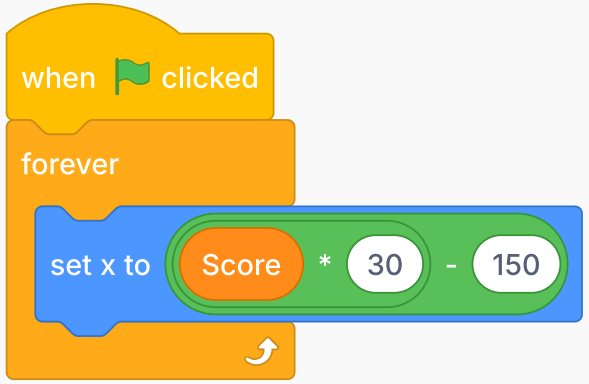
If it doesn't work

Every answer is wrong: the + block needs **A** and **B** dropped in.

Wrong answers say Correct!: the second if needs the **not**.

Missions 5 and 6: walk and win

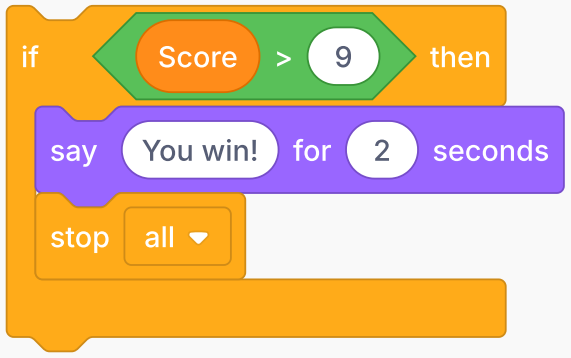
MISSION 5 · A SECOND SCRIPT ON THE CAT



SCORE	SCORE * 30	- 150	THE CAT IS
0	0	-150	on the left
5	150	0	in the middle
10	300	150	on the right

*	times: 30 steps for every point.
-	take away 150, so the cat starts on the left.
>	more than 9 means 10 right: you win.

MISSION 6 · AT THE BOTTOM OF THE FIRST SCRIPT'S FOREVER



If it doesn't work

The cat doesn't walk: put Score inside the *, and the * **inside** the -.

It never ends: the win check goes inside the forever, after the other two ifs.

Save your game

1. Click **File**, then **Save to your computer**.
2. Name it **Math Cat** and your name.
3. Close the Scratch tab.

> Please don't show the next group your game. They're taking the same test.